Web Appendix

Do Influencers Pay? Limited Attention, Engagement Elasticity, and Simulation-Based Pre-Tests for Seeding Decisions

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Web Appendix A: Simulation Settings and Assumptions

Table W1. Simulation Parameters and Assumptions

Parameter	Value	Remarks
Stimulus scale κ	0.16	Set manually
Stimulus decay τ	1 hour	Single exponential
Stimulus cap S_{max}	3.0	Synthetic saturation
Base exposure probability	0.5	Multiplied by $(\max\{f, 10\}/10)^{-\beta}$
Engagement elasticity β	0 (base); 0 to 0.66	Applied to all accounts
Comment visibility factor	0.4	Multiplies exposure probability
Feed capacity	40 items	Most recent items
Recency weight	exp(−age/2 hours)	For comment and retweet targets
Circadian profile	0.12 + evening Gaussian at chronotype (SD 2.2 hours) + 0.45 × midday Gaussian at 12.5 hours (SD 1.8 hours)	Normalized to mean one; 15-minute resolution
Emotion at each urge evaluation	excited / angry / happy / indifferent with probabilities .20 / .15 / .30 / .35	Assumption
Relevance level	interest < 0.10: 0.25; < 0.30: 0.50; otherwise 0.75	Assumption
Network	2,000 agents, preferential attachment, 3 follows per new agent	
Horizon	18-hour warm-up; 48 hours after injection	Injection at 18:00 unless stated
Distributed seed pool	Accounts outside the top 5% by followers	Random order until the budget is spent
Volume-matching factor	2.00	Alternative schedulers only; ratio of total events in pilot runs without a campaign
Seed price floor	One follower-equivalent	Accounts without followers are priced as one follower
Adoption	First comment or retweet of campaign content	Seeds are not counted as adopters

Notes: Likes create no items and no notifications. The code in the supplementary archive reproduces every value.

Rule-based urge function. With h the hour of day, leisure = 1 if $h \geq 19$ or $h < 1$, 0.5 if $11 \leq h < 14$, and 0 otherwise; n the number of feed items in the last hour on topics with interest weight ≥ 0.10; $\ell \in \{0, 1, 2\}$ the relevance level of the latest item; $\nu = 1$ for a notification and 0 otherwise; and e an emotion bonus of 0.8 (excited), 0.6 (angry), 0.4 (happy), or −0.6 (indifferent), the rating is $I = \min(\max(1.0 + 1.0 \times \text{leisure} + 0.4 \times \min(n, 3) + 0.5 \times \ell + 0.8 \times \nu + e, 0), 5)$. For notifications ℓ is set to 1.

Web Appendix B: Validation of the Estimation Procedure on Synthetic Data

To check that the procedure in the main text recovers known parameters, we generated synthetic campaign data with the structure of an influencer-campaign dataset with timestamps: 400 authors with follower counts drawn log-uniformly between 10^5 and 10^7, assigned to six products, 15 posts per author over 60 days, author and product random effects (SD 0.8 and 0.3), post-level noise (SD 0.5), log-normal reply delays (median 0.8 hours, log-SD 1.2), and interaction times thinned by a known daily profile with an evening peak at 21:30. Table W2 reports the results for two true values of β. The random-effect scales are our choices, because the structure rather than the scale is what the exercise needs.

Table W2. Poisson Pseudo-Maximum Likelihood Recovers the True Elasticity, Whereas OLS on the Log Count Does Not

True β	PPML β̂ (SE) [95% CI]	OLS on ln(1 + y) β̂	Fitted evening peak	Held-out TV (flat profile)	Held-out r
0	-0.064 (0.064) [-0.189, 0.061]	-0.003	21.6 hours	0.034 (0.286)	0.992
0.35	0.346 (0.042) [0.264, 0.428]	0.829	21.5 hours	0.046 (0.317)	0.991

Notes: 6,000 posts per row. PPML = Poisson pseudo-maximum likelihood; SE = standard error implied by the confidence interval (clustered by author). TV = total variation distance between the fitted profile from three calibration products and the observed hourly distribution in three held-out products; the value for a flat profile is shown in parentheses. The fitted reply-delay median was 0.80 and 0.79 hours in the two cases, against a generating value of 0.8 hours.

The PPML confidence interval contained the true β in both cases. OLS on the log-transformed count was close to the truth when the true β was 0 but far off when it was 0.35 (0.83), because adding one to the count distorts the relation between interactions and audience

size. The daily profile fitted on calibration products reproduced the held-out profile closely in both cases ($r = 0.99$), far more closely than a flat profile. The synthetic data include posts with no interactions and only accounts with 10^5 to 10^7 followers, so this exercise does not test the estimator under the zero truncation and account selection present in SAGraph, which the main text addresses with the zero-truncated model and a discussion of bias. Because the public SAGraph release lacks timestamps, only the elasticity part of the procedure could be applied to real data.

Web Appendix C: Additional Estimates of the Engagement Elasticity

Table W3. Elasticity Estimates Are Similar Across Robustness Checks, but Imprecise for the Largest Accounts Within a Campaign

Sample	Posts	Authors	$\hat{\beta}$ PPML (SE) [95% CI]	$\hat{\beta}$ ZTP (SE) [95% CI]
Pooled, $10 \le f < 10^3$ (selected sample; not interpreted)	50,342	21,580	1.02 (0.023) [0.98, 1.07]	1.03 (0.029) [0.97, 1.09]
abc_reading, $f \ge 10^5$	28,574	1,415	0.56 (0.099) [0.37, 0.76]	0.56 (0.099) [0.37, 0.76]
electric_toothbrush, $f \ge 10^5$	1,556	180	0.51 (0.180) [0.16, 0.86]	0.51 (0.180) [0.16, 0.86]
intelligent_floor_scrubber, $f \ge 10^5$	23,256	1,276	0.59 (0.088) [0.42, 0.77]	0.59 (0.088) [0.42, 0.77]
ruby_face_cream, $f \ge 10^5$	1,826	135	0.36 (0.143) [0.08, 0.64]	0.36 (0.143) [0.08, 0.64]
spark_thinking, $f \ge 10^5$	27,889	1,416	0.60 (0.066) [0.47, 0.72]	0.60 (0.066) [0.47, 0.72]
supor_boosted_showerhead, $f \ge 10^5$	3,487	517	0.65 (0.122) [0.41, 0.89]	0.65 (0.122) [0.41, 0.89]
Pooled, comments only ($f \ge 10$)	209,178	38,279	0.74 (0.009) [0.72, 0.76]	0.72 (0.010) [0.70, 0.74]
Pooled, all accounts with f set to at least 10	219,379	43,077	0.72 (0.019) [0.68, 0.75]	0.70 (0.022) [0.66, 0.74]
Pooled, clusters merged across campaigns	209,890	38,279	0.66 (0.015) [0.63, 0.68]	0.63 (0.017) [0.59, 0.66]

Notes: Standard errors (SE) are clustered by author within campaign, except in the last row, where accounts that appear in several campaigns (identical description and following count, same order of magnitude of followers) share one cluster. A post is a unique pair of author and post text; the dependent variable counts comments and reposts, except in the comments-only row. For accounts with 10 to 10^3 followers the estimate is close to 1, but such accounts enter the data only when one of their posts received an interaction, which inflates their measured

engagement; we do not interpret this estimate. PPML = Poisson pseudo-maximum likelihood; ZTP = zero-truncated Poisson.